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Unraveling Electronic Trends in O* and OH* Surface Adsorption in the MO_2 Transition-Metal Oxide Series

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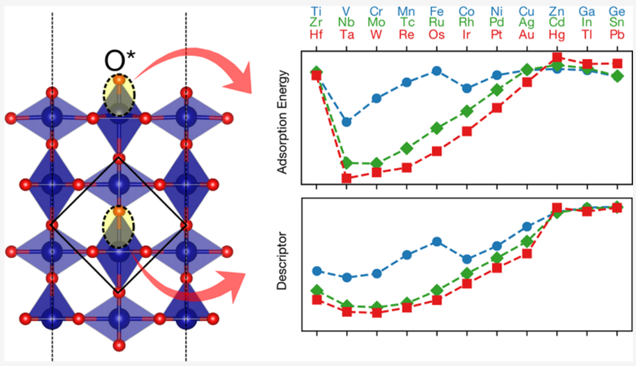
ABSTRACT: Understanding the bond strength of O* and OH* intermediates to metal-oxide surfaces is key to predicting the catalytic activity in oxygen-based electrochemistry. In this work, we uncover highly non-linear trends in O* and OH* adsorption energies across the 3d, 4d, and 5d series of MO_2 transition-metal (TM) oxide surfaces computed within Hubbard-U corrected density functional theory (DFT + U). Investigating the electronic structure with crystal orbital Hamiltonian populations (COHP) of the relevant metal−oxygen bonds reveals that the spin-dependent coupling strength between metal-d and oxygen-2p atomic orbitals together with the extent of filling of bonding and anti-bonding orbitals are the primary contributors to the adsorption energy. Importantly, we show that the integrated COHP obtained purely from bulk calculations is a highly accurate descriptor for surface adsorption energetics that captures trends across the group 5−12 TM oxide series within 0.19−0.36 eV. Our results suggest a pathway to prediction of adsorption energies for an arbitrary metal−ligand catalyst system.



■ INTRODUCTION

Producing high-performing and earth-abundant catalysts for the oxygen evolution and oxygen reduction reactions is important for clean hydrogen production from water and the creation of viable fuel cells.[1−5] For these reactions, O* and OH* intermediates have been demonstrated to be key descriptors for predicting the catalyst activity.[1,6,7] While much work has been done to understand the relative binding energies of these two species, the overall trends for oxide materials[1,3] and their underlying descriptors[8−10] are much less understood than for metals and metal alloys.[2] More generally, reactivity patterns in metal oxides across d-electron configurations have a long standing interest to both the experiment[11−14] and theory.[8,15−17]

In this work, we examine the adsorption energetics of O* and OH* on metal oxide materials in the MO_2 rutile-type crystal structure. We select the +4 oxidation state as the majority (27 of the 33) of the studied metals have been observed in this oxidation state according to the Inorganic Crystal Structure Database,[20,21] the rutile structure being a common polymorph (19 of the 33). This selection allows us to examine the entire 3d, 4d, and 5d series and focus on variations in chemical and electronic composition rather than structural arrangement. Previous works focusing on such electronic effects have shown near−linear correlation between adsorption energies of O* and OH* and the number of outer d-electrons for a given oxidation state.[8,22,23] However, these correlations did not examine the deviations of group 4 and 12−14 elements, differences in adsorption strength along the 3d, 4d, and 5d periods as well as notable spin transition effects. The O(2p) band center of the adsorbed oxygen[9] was demonstrated to correlate with adsorption energy of O* and OH* across many 4d and 5d transition metals and metallic oxides but was not shown to have predictive power beyond these systems. Recently, Fung et al.[10] showed a good correlation between the integrated COHP of the bulk metal−oxygen bond and the energy of adsorption of hydrogen onto lattice surface oxygen sites in cubic perovskites.

Here, we demonstrate that the trends in O* and OH* adsorption are captured by the integrated COHP with a very high accuracy across the whole transition-metal (TM) rutile series, including where strong spin and crystal-field effects are present. We identify notable bonding extrema in each series and describe the reactivity patterns in OER across all d-electron configurations. In particular, we apply the COHP analysis to the metal d and oxygen 2p orbitals of the metal−oxygen bond for both the surface and the bulk and show that

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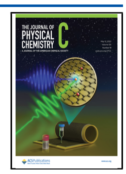

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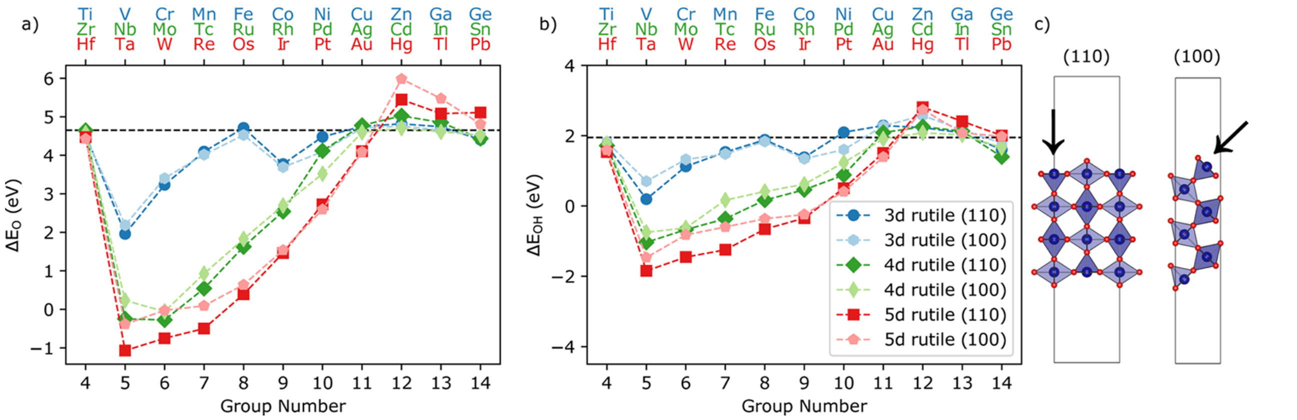

Figure 1. Adsorption energies of (a) O* and (b) OH* on the rutile (110) and (100) surfaces as a function of group number of the periodic table. The dotted line indicates the approximate weak binding limit. (c) Cross section of the empty site (*) for (110) and (100) surface geometries used in this work, where arrows indicate the active site. Data is made available at Catalysis-Hub.org.[18,19]

in both cases, the COHP is a strong descriptor for adsorption energetics.

METHODS

All DFT calculations were performed using the Viena ab initio software package (VASP)[24−29] using the PBE + U method. A plane wave cutoff of 500 eV was selected, and a k-point grid of $8 \times 4 \times 1$ was used for slab calculations. All relaxations were performed to an accuracy of 0.02 eV/Å except for oxygen adsorbed to the CdO_2 (100) surface which was converged to an accuracy of 0.07 eV/Å. All calculations were done with spin polarization. In addition to structural and electronic optimization, spin degrees of freedom were also fully relaxed under the condition that empty*, OH*, and O* surfaces share the same magnetic structure (see Figure S1 for illustration of this principle for CrO_2). Hubbard U values were applied for all 3d metals with an open d-shell to compensate for the over-delocalization of d electrons common in the GGA methodology, and 4d and 5d metals were treated at the PBE level. The Hubbard U values are tabulated in the Supporting Information, Table S1.

COHP calculations were performed using the Lobster package,[30] starting from re-converging the electronic structure with the symmetry disabled and the k-point grid set to a finer grid of $12 \times 6 \times 1$. The COHP between metal d-orbitals and oxygen 2p-orbitals was investigated, including the metal−oxygen bond of the bulk geometry as well as the surface bond between the adsorbed oxygen and the metal site.

RESULTS AND DISCUSSION

O* and OH* Adsorption on Rutile Oxides. The calculated adsorption energies of O* and OH* on MO_2 rutile surfaces are shown in Figure 1a,b. Because the formal oxidation state of the metal is +4, we span groups 4−14 corresponding to d^0 to d^{10} configurations for 3d, 4d, and 5d rows in the periodic table. Adsorption energies are taken relative to the gas phase of molecular water and hydrogen. We consider CUS-site adsorption on (110) and top site at the (100) surface at high coverage, in their respective 1×1 cells pictured in Figure 1c.

Overall, adsorption energies are seen to evolve in a progressive, non-monotonic fashion both as a function of group number, determining the number of outer d-electrons,[8,23] and d-period (3d, 4d, 5d row). In more detail, all group 4 TM oxides are weakly binding, and a sharp increase in adsorption strength occurs from group 4−5 forming well-defined minima for V, Nb, and Ta. This is followed by a linear weakening in adsorption with increasing group number until group 12, followed by slight strengthening in adsorption due to interaction with metal p-states in groups 13 and 14. In the limits of d^0 or d^{10} configurations (group 4 and 12−14), where the d orbitals are unlikely to contribute to additional bonding, a weak binding limit of 4.65 and 1.95 eV is observed for O* and OH*, respectively.

While the adsorption energy curves for 4d and 5d metal oxides show broadly the same form, notably, an extra minimum is formed for 3d Co^{4+} with a sharp discontinuity from Fe to Co. This is a direct consequence of transition from high to low spin on the transition metal site, where V, Cr, Mn, and Fe takes a high spin state, with a monotonically increasing magnetic moment until reaching Fe, as shown in Figure S2. High spin states are frequently observed for 3d transition metal complexes as explained by crystal field theory.[31] Interestingly, we find that the high spin on V, Cr, Mn, and Fe leads to weaker adsorption.

The resulting double minima in the 3d adsorption leads directly to Dowden's double-peak behavior the OER activity for the 3d oxides (see Supporting Information Figure S3), with peaks in activity around VO_2−CrO_2 (group 5−6) as well as for CoO_2 (group 9), previously discussed for other reactions on metal-oxides.[11,12,17] We also show that due to lack of spin transitions, 4d and 5d metal oxides can only have a single activity peak. To further underline the universality of adsorption energy trends of +4 3d TM systems, we demonstrate that $SrMO_3$ perovskites, layered MO_2,[32] and rutile (110) and (100) all show similar behavior (see Figure S4).

The variation in adsorption energy between (110) and (100) surface facets is relatively small, which we attribute to the structural similarity of the octahedral coordination environment of the active sites, which is further supported by non-rutile data shown in Figure S4. The formal oxidation of the active site varies slightly between the facets; +3.33 and +4.0 for the (110) and (100) surface orientation, respectively. Here, the site-specific formal oxidation state is calculated from the metal−oxygen coordination, assuming charge neutrality and that all oxygen atoms take a formal charge of −2. In this scheme, the adsorption of oxygen to the metal top site leads to

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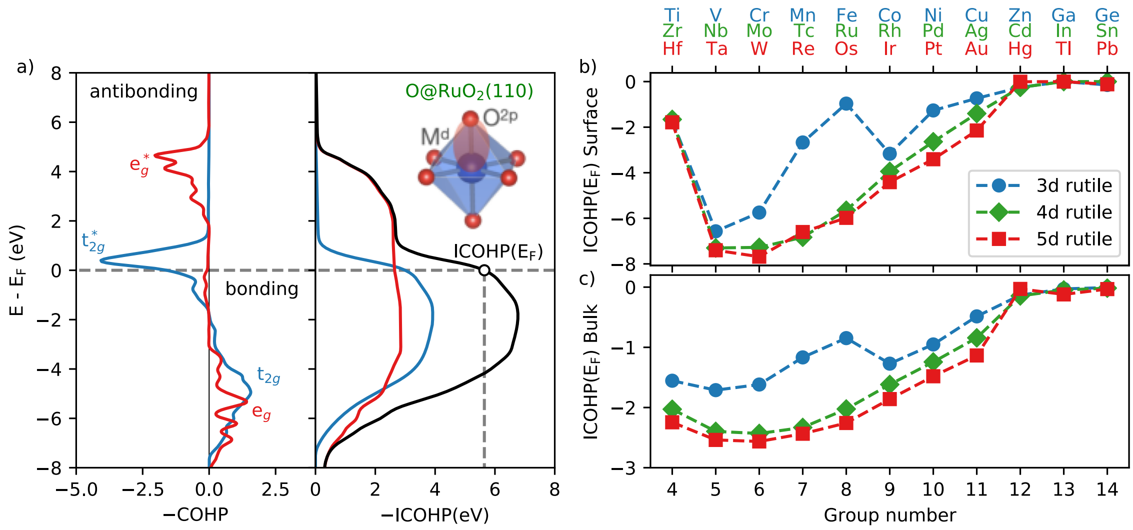

Figure 2. (a) O(2p)−Ru(4d) COHP for O* adsorbed on the rutile-structured RuO_2 (110) surface. The left portion shows the normal COHP, and the right shows the integrated COHP. The blue and red curve represents the t_{2g} and e_g d-orbital components, respectively, and the black curve represents the sum of the two (total interaction of the Ru 4d-orbitals with the O 2p-orbitals). Trends in the (b) surface ICOHP(E_F) and (c) bulk ICOHP(E_F) showing remarkable resemblance to the adsorption energy trends in Figure 1.

an idealized increase of +2 in oxidation state, compared to a +1 increase for OH adsorption. We note that the formal oxidation state only corresponds to the actual oxidation state in the limit of ionic bonding and that a significant covalent character is present for the systems in question.

COHP as a Qualitative Electronic Descriptor. While the effects described above may seem to dispute a model with a single descriptor, we have found that the integrated crystal orbital Hamiltonian populations[33] (ICOHP) does an excellent job of capturing these trends. COHP is a method for studying the bonding and anti-bonding interactions within a chemical system.[34,35] In this work, we utilize the LOBSTER package to perform COHP calculations.[30,36] The COHP method involves projecting the plane wave basis sets solution obtained from ground-state DFT solution (using the VASP code[24−29]) into LCAO basis functions. From this, the Hamiltonian between two projected orbitals can be calculated, indicating if the interaction is net bonding or anti-bonding at a particular band energy.[33,37] Thus, the COHP can be thought of as the energy contribution arising from the overlap of two orbitals. This energy contribution can be either bonding (negative) or anti-bonding (positive).

Figure 2a provides a representative COHP plot, showing the energy-resolved COHP between the 2p-orbitals of adsorbed oxygen and a surface Ru d-orbital for rutile-structured RuO_2. The negative COHP is plotted following the convention in the literature, such that bonding interactions correspond to positive values. The interaction between oxygen 2p-orbitals and Ru d-orbitals has been split into the e_g and t_{2g} symmetries, relevant for the octahedral coordination environment that corresponds to head-on interactions (σ bonds) and side-on interactions (π bonds), respectively. From this breakdown, we can see that below the Fermi level, both e_g and t_{2g} form bonding interactions that are completely filled. However, near the Fermi level, the Ru t_{2g} orbitals form an anti-bonding interaction, weakening the bond. As one moves across the row, the states move below the Fermi level filling antibonding levels and thereby weakening the bond.

In Figure 2a, we show the cumulative integral, referred to as the ICOHP. The ICOHP corresponds to the energy gained from the pair-wise bonding interaction, analogous to the integrated DOS yielding the number of electronic states. The Ru−O ICOHP was seen to increase with the filling of bonding orbitals, and value of the Fermi level corresponds to integrating over all occupied states and can be thought of as a measure of the bond energy. At high energies, both bonding and antibonding orbitals are filled, resulting in net-zero bonding.

Next, we consider the relationship between the ICOHP(E_F) and the adsorption energy. In Figure 2b, we show the ICOHP(E_F) between the metal d-orbital states of the adsorption site and the 2p states of the adsorbed oxygen for the full series. The trend qualitatively mirrors that of the adsorption energy of the oxygen (Figure 1a) with a large ICOHP corresponding to strong adsorption. In Figure 2c, we show the ICOHP extracted from the metal−oxygen pair of the bulk calculations. The bulk ICOHP show very similar trends to the surface COHP and the adsorption energies with the exception of the group 4 elements (Ti, Zr, and Hf) that will be discussed later.

ICOHP as a Quantitative Electronic Descriptor. To examine the ability of ICOHP to quantitatively model adsorption energetics, we plot the ICOHP(E_F) against the adsorption energies obtained from DFT. Figure 3a shows the relationship between the surface-derived ICOHP and the adsorption energies of O*/OH* on the (110) rutile surfaces; the result for (100) rutile is shown in Figure S5. In the strong adsorption regime, the correlation between the ICOHP and the adsorption energies is approximately linear but in order to capture the behavior in the entire adsorption energy range, a quadratic fit is needed. While the surface-derived ICOHP is highly correlated with the adsorption energy, fairly large deviations are found for CrO_2 and VO_2, leading to RMSE/MAE values of 0.74/0.47 and 0.43/0.37 eV for O* and OH*, respectively. The deviation can be explained by structural differences between the empty and occupied adsorption site that cannot be captured by the metal-adsorbate oxygen bond alone. The restructuring is more dominant for oxygen adsorption in group 5 and 6 where we also observe the largest deviation. CrO_2 and VO_2 have been found to undergo metal−insulator transitions[38] which is consistent with the observation

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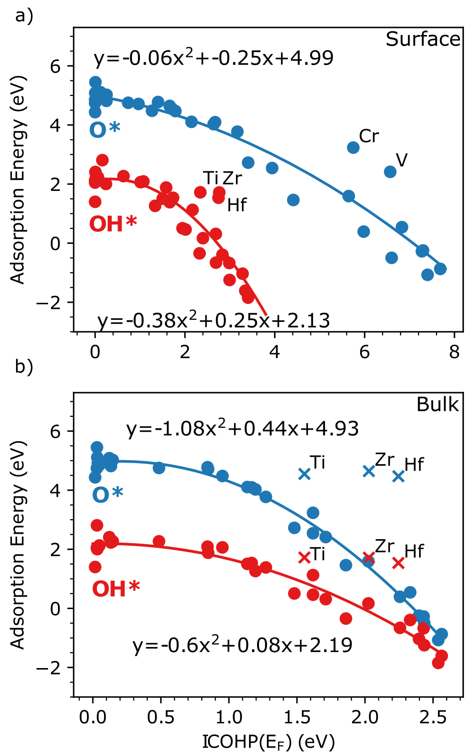

Figure 3. (a) Correlation between adsorption energy and surface ICOHP with R^2 values of 0.91 and 0.77 and RMSE/MAE values of 0.74/0.47 and 0.43/0.37 eV for O* and OH*, respectively, (b) adsorption energy and bulk ICOHP correlation with R^2 values of 0.99 and 0.94 and RMSE/MAE values of 0.19/0.15 and 0.21/0.16 eV. The group 4 oxides (x points) have been discarded from the fit in (b).

of a larger driving force for structural reorganization in these systems. We note that the surface ICOHP is non-practical for screening purposes, since the computational cost needed to obtain the ICOHP is comparable to that of calculating the adsorption energy.

Surprisingly, the bulk ICOHP shows a superior correlation with adsorption energetics. As shown in Figure 3b, the bulk ICOHP successfully captures the variation over the 3d, 4d, and 5d transition-metal oxide series, including the eccentric behavior of the 3d metal oxides caused by spin effects. Interestingly, the adsorption on CrO_2 and VO_2 surfaces are well captured in spite of structural reorganization of the surface site, indicating that the bulk ICOHP better captures the equilibrium bond strength. The only large deviation is the group 4 elements (Ti, Hf, and Zr), where the bulk COHP significantly overestimates the binding strength of the adsorbate. When excluding the group 4 points from the quadratic fit, we obtain small RMSE/MAE values of 0.19/0.15 and 0.21/0.16 eV for O* and OH*, respectively. The primary reason for the discrepancy for the group 4 oxides is a mismatch between the formal oxidation state of the surface (+5 to +6 after adsorption) and the bulk (+4). Although the actual oxidation state is unlikely to surpass +4 for many of the oxides considered here, a larger formal oxidation state of the surface generally reflects an increased metal−oxygen coordination. This implies that the surface metal atom will contribute a smaller electron density to each surrounding metal−oxygen bond. This leads to a decrease in bond strengths for group 4 oxides, where only four valence electrons are available. This highlights the importance of selecting bulk states that closely match the filling conditions on the surface or introduce further correction schemes to capture the filling state of the surface.

The relationship we find between the ICOHP and O* and OH* adsorption is similar to the one seen by Fung et al.[10] for H adsorption, where the bulk metal−oxygen ICOHP was shown to capture the bonding of H to a (bulk-like) surface lattice oxygen on cubic perovskites. In this work, we show that the bulk ICOHP descriptor is applicable for the prediction of adsorption energetics of geometries with significantly larger

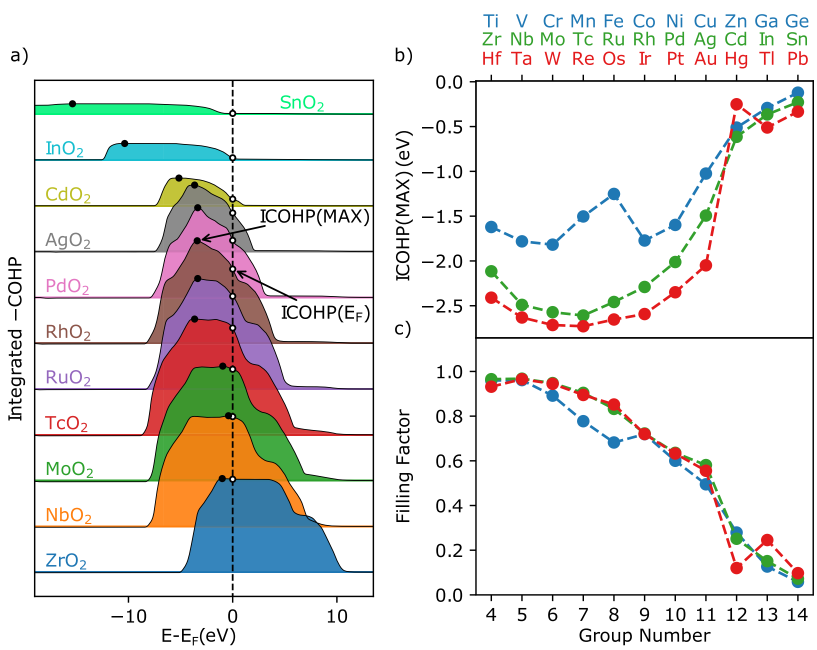

Figure 4. (a) ICOHP curves for the 4d bulk rutile oxide series, where curves for 3d—5d are shown in Figure S6. (b) Maximum ICOHP value for each material studied (the black points from a). (c) Filling factor obtained as the ICOHP(E_F)/ICOHP(MAX) ratio, where the ICOHP(E_F) is shown in Figure 2c.

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structural freedom including changes to the metal−oxygen bond length of the adsorbate. Furthermore, we have shown that the ICOHP is able to account for the complicated spin effects in the 3d metals. Given the strong correlation between the ICOHP and the adsorption energies of H*, O*, and OH* on perovskite and rutile oxide surfaces, we believe this method has promise for a broader range of adsorbates and materials.

Insights into Electronic Structure from COHP. In the following, we demonstrate that the COHP analysis also leads to a deeper understanding of the non-trivial evolution of the adsorption energy trends across the transition-metal oxide series. In Figure 4a, the full ICOHP curves are shown for the 4d series of the bulk rutiles, and the plots for the full 3d, 4d, and 5d series are provided in Figure S6. The ICOHP curves are seen to evolve in a similar manner, the value increasing and decreasing with the filling of bonding and antibonding states, respectively. The value at the Fermi level, ICOHP(E_F), is marked with white circles. The maximum value, ICOHP(MAX), where only bonding states are filled, are marked with black circles. In all cases, the bonding states are filled, while only for group 4 (TiO_2, ZrO_2, and HfO_2), the antibonding states are completely empty. With the increased filling of antibonding states, the ICOHP curves shift to the left resulting in smaller ICOHP(E_F). We also see that a decrease in orbital filling, as expected for the surface-adsorbate bond, will result in a shift of the Fermi level to the left and a weakening of the bond for group 4 oxides. This is consistent with the observed discrepancy between the bulk ICOHP and the adsorption energies for the group 4 oxides.

Interestingly, with increased filling, the ICOHP curves also decreases in width and magnitude. Thus, we see that the variation of the ICOHP(E_F) depends on two quantities: (1) the variation in the maximum value, ICOHP(MAX), and (2) the effect of filling anti-bonding orbitals. The variation in ICOHP(MAX) for the entire series can be seen in Figure 4b. We immediately see that changes in bond strength between the 3d, 4d, and 5d series correlates with ICOHP(MAX) that decrease in magnitude going from 3d to 4d and 5d oxides. On the other hand, the ICOHP(E_F)/ICOHP(MAX) ratio, termed Filling factor in Figure 4c, is almost identical for the 4d and 5d oxides, decreasing from 1 to 0 in a monotonic fashion with increased filling. The Filling factor for the 3d oxides shows similar behavior, although small deviations are observed for CrO_2, MnO_2 and FeO_2 due to a splitting of spin up and down states.

The trend in ICOHP(MAX) can be further understood in terms of a metal-dependent orbital coupling, driven by an increased localization of 3d-orbitals compared to 4d and 5d.[39] This effect has been demonstrated for adsorption on pure transition metals in the context of the d-band theory, where the overlap matrix element of the Hamiltonian (V_{ad}) between a transition metal surface and the adsorbate is an important quantity.[40] The COHP provides a similar measure that is formally expressed as the product of the Hamiltonian matrix element and the density matrix. Interestingly, the bulk ICOHP(MAX) captures the spin-dependent reduction in binding strength for Cr, Mn, and Fe, thus indicating a spin-driven reduction in orbital coupling.

There are several benefits to using ICOHP as a descriptor for adsorption energies, the largest being that ICOHP obtained from relevant bulk materials allows modeling of surface adsorption at significantly lower computational cost than surface calculations. Although the relevance of the bulk-derived descriptor to surface quantities is not guaranteed, we believe that the bulk ICOHP is able to capture the basic nature of metal−oxygen bonding which remains roughly the same on the surface. Furthermore, separating the effects of orbital filling and matrix overlap will enable surface-facet specific filling corrections based on the oxidation state of surfaces. We envision that facet-dependent filling corrections will enable the prediction of different facets and oxidation states. Further work will be needed to extend the ICOHP descriptor to predict adsorption on higher miller index facets and under-coordinated sites. The ICOHP analysis also provides direct probing of bonding and anti-boding interactions rather than orbital populations as is the case for DOS calculations. Here, separating the effects of orbital filling and matrix overlap has the potential to enable surface-facet specific filling correction to further generalize the model.

■ CONCLUSIONS

In conclusion, we have demonstrated the usefulness of bulk ICOHP calculations in the prediction of O* and OH* adsorption on oxide materials. We note that the descriptor is able to capture the bonding relationship across 3d, 4d, and 5d with an MAE of roughly 0.19−0.36 eV on rutile surfaces−accurate enough to screen materials for their approximate position on a volcano plot. We demonstrated that the ability of COHP to capture this behavior is related to two factors: the coupling of the metal and oxygen orbitals and the filling of bonding and anti-bonding states. Due to the success in predicting H*, O*, and OH* adsorption, we believe that the ICOHP descriptor will be relevant for machine-learned model of surface adsorption from bulk calculations.

■ DATA AVAILABILITY

The adsorption energy data set including surface slabs, gas phase, and bulk geometries are made available at http://www.catalysis-hub.org/publications/ComerUnraveling2022.[18,19] The calculated COHP and ICOHP spectra are made available on github.com.[41]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcc.2c02381.

> Additional computation details, including values for Hubbard *U* corrections and converged magnetic moments, calculated OER overpotentials for the rutile set, supporting adsorption energy calculations for other 3d +4 transition-metal oxide structures, COHP versus adsorption energy correlation for the rutile (100) facet, and full ICOHP curves for the 3d, 4d, and 5d oxides, and tabulated DFT adsorption enthalpies and free energies (PDF)

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Notes

The authors declare no competing financial interest.

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